

Calibration of Production Transient Alert Brokers: Preparing Decision Infrastructure for LSST

Can broker confidence scores be trusted for automated follow-up decisions?

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1 Motivation

“When a broker reports 80% confidence, is it actually correct 80% of the time?”

Millions of alerts → Broker probabilities → Gates & rankings →

Limited follow-up

- Rubin/LSST alert volumes cannot be inspected manually.
- A classifier can rank the correct class first while still reporting an unreliable confidence.

Underconfidence

Correct predictions score too low → useful objects fail fixed gates.

Overconfidence

Wrong predictions score too high → unreliable labels auto-accepted.

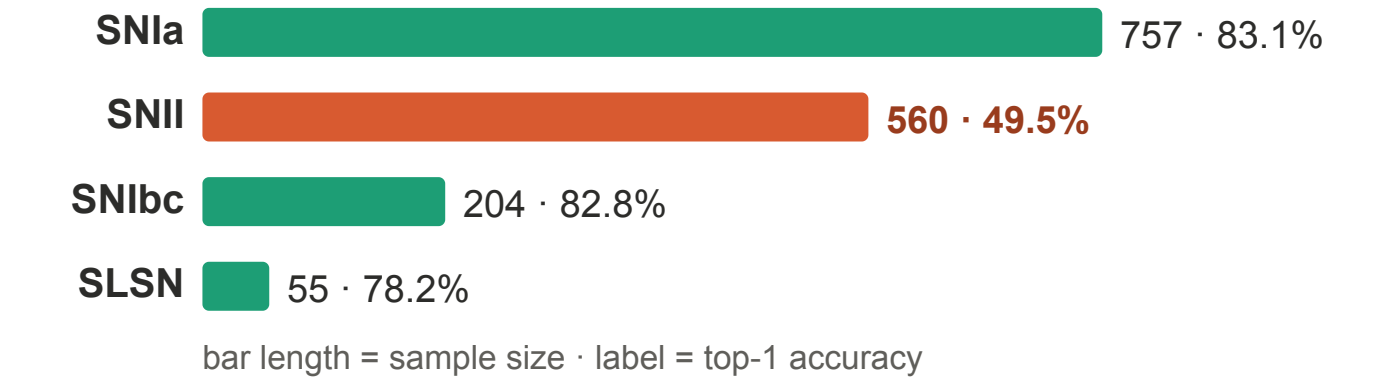
A functioning classifier is not automatically functioning decision infrastructure.

2 Data & ground truth

Spectroscopically confirmed ZTF Bright Transient Survey objects, cross-matched against the deployed outputs of two production classification systems. BTS spectroscopy is the independent ground truth.

System	Role	Task
ALeRCE	production alert broker	15-class
NEEDLE	rare-transient classifier (not a broker)	3-class

ALeRCE class counts and accuracy



Overall ALeRCE accuracy 70.9%, but strongly class-dependent — SNII is the classification bottleneck.

PROPOSED Calibration-aware spectroscopic deferral

Broker probabilities
↓
Calibration layer
↓
Checker risk $r_i = P(\text{classification is wrong})$
↓
Scientific value & observability
↓

Top-B targets for spectroscopy

$$\max_z \sum_i z_i t_i V_i q_i \quad \text{s.t.} \quad \sum_i z_i c_i \leq B$$

z_i select for spectroscopy · r_i checker error risk · V_i scientific value · q_i probability of a usable spectrum · c_i exposure cost · B observing budget. In words: choose the candidates where a spectrum buys the most science within the available budget. A separate novelty channel will flag objects outside the broker taxonomy.

Not implemented or tested here — these results motivate a future calibration-aware learning-to-defer framework.

3 Method

1 Score correctness. Top-ranked class vs independent spectroscopic label.

2 Audit calibration. Does reported confidence match observed accuracy?

$$\text{ECE} = \sum_b (n_b/N) \cdot | \text{acc}(b) - \text{conf}(b) |$$

3 Recalibrate. Held-out temperature scaling. Brokers expose probabilities, so log-probabilities act as pseudo-logits and softmax returns a valid simplex.

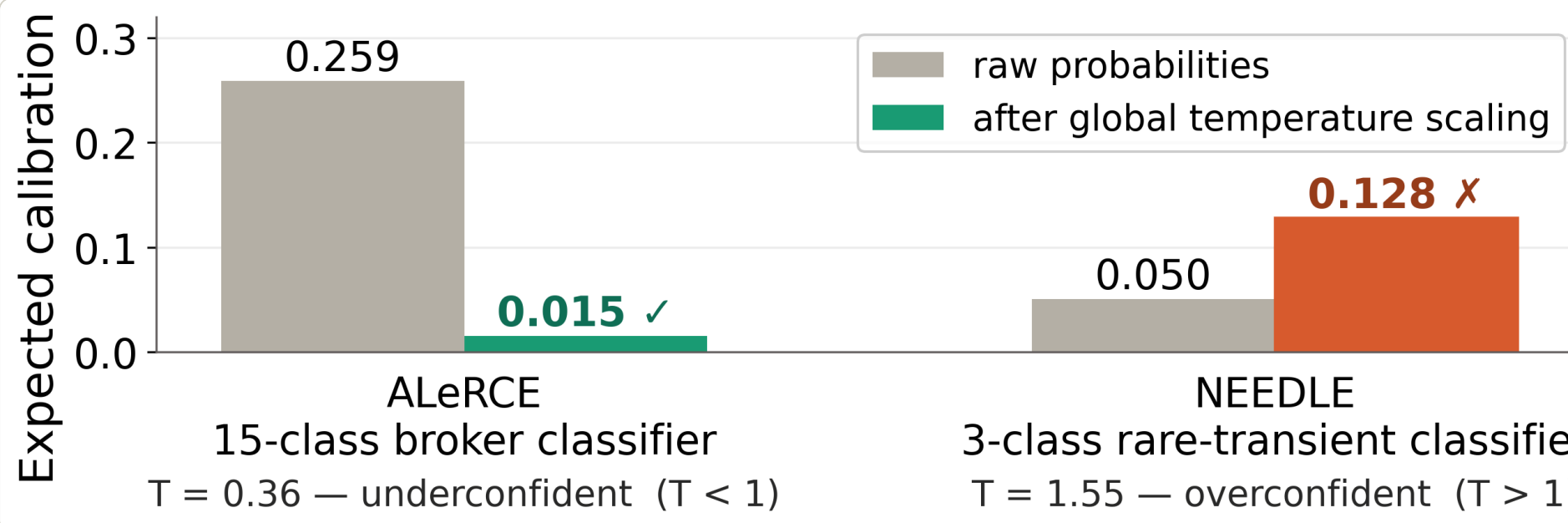
$$\tilde{p} = \text{softmax}(\log p / T) \quad \text{monotonic} \rightarrow \text{top-1 class unchanged}$$

4 Train a reliability checker. Logistic model estimating P(top-1 correct) from the calibrated vector plus its shape (margin, entropy, branch mass), benchmarked against a predicted-class-rate control.

5 Accept or defer. Rank by estimated reliability; defer the least trustworthy fraction.

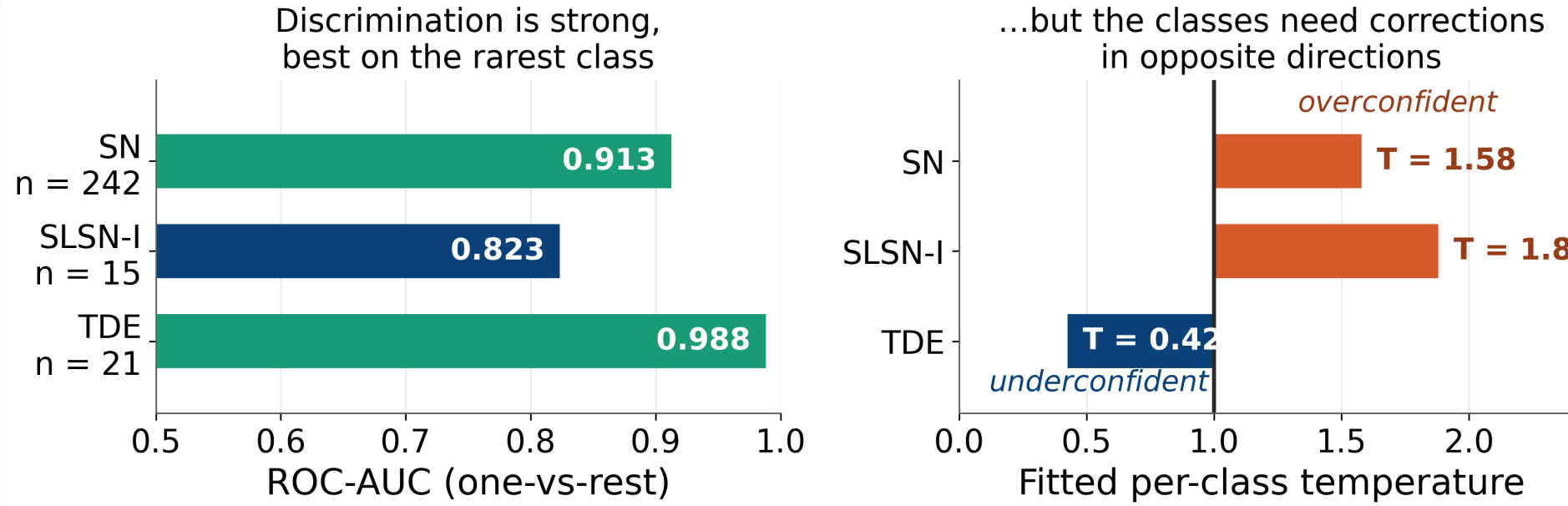
Five-fold out-of-fold · paired stratified bootstrap · ROC-AUC, AURC and risk-coverage. Deferral means more photometry or human review — not automatically spectroscopy.

4 Two systems, opposite failure modes



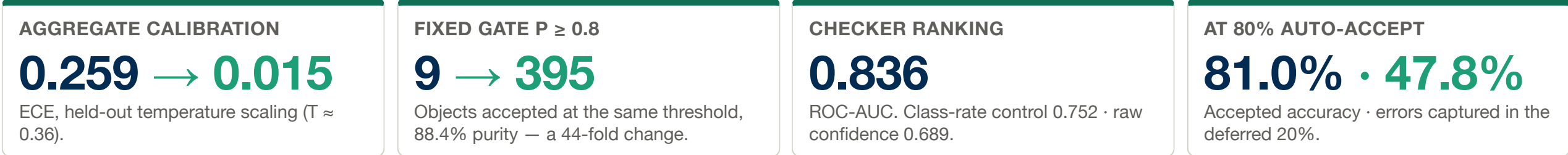
ALeRCE is underconfident and is repaired. NEEDLE starts far better in aggregate but is overconfident, so the identical procedure makes it worse. Aggregate ECE does not tell you which correction a system needs.

NEEDLE per class: strong ranking, conflicting calibration

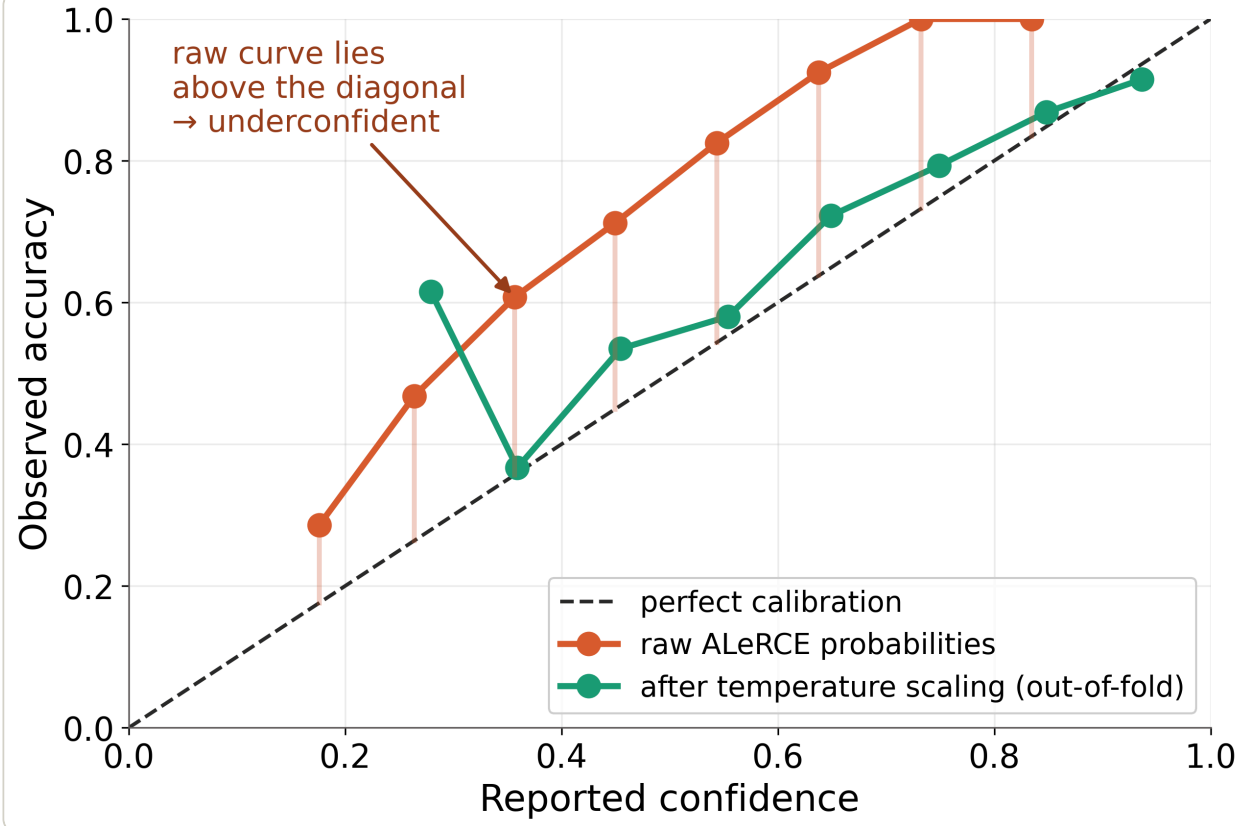


NEEDLE ranks its classes well — best on its rarest, TDE (ROC-AUC 0.988, 21/21 recovered). But its classes need corrections in opposite directions, so no single temperature serves all three — which is why the global fix backfires. Rare-class samples are small: n = 15, 21.

5 Results — repairing and using the ALeRCE scale

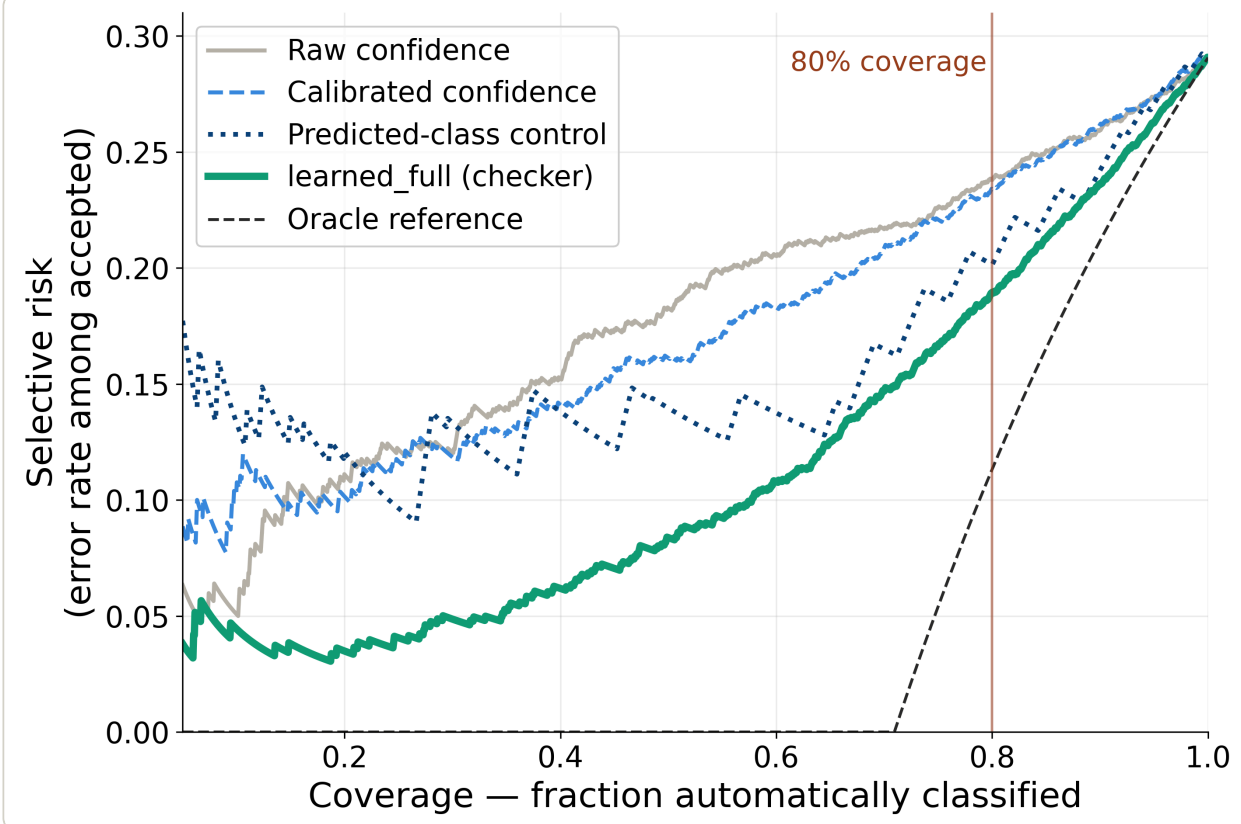


ALeRCE is systematically underconfident



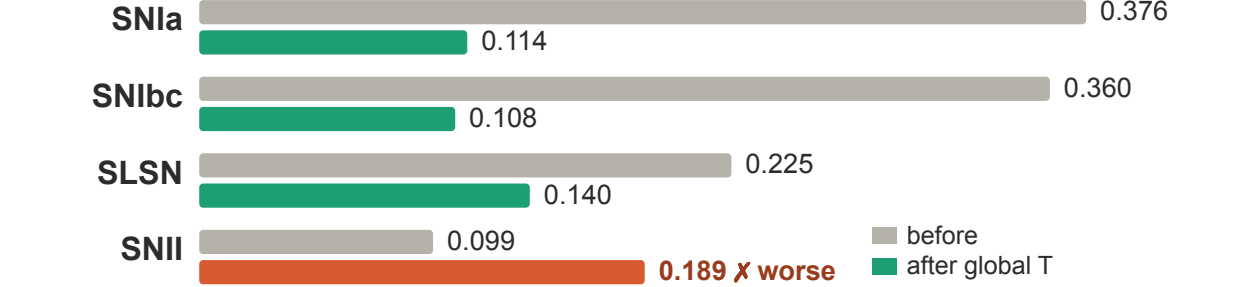
Raw ALeRCE sits far above the diagonal — correct far more often than it claims. Temperature scaling pulls it onto the diagonal without changing any predicted label.

Selective classification



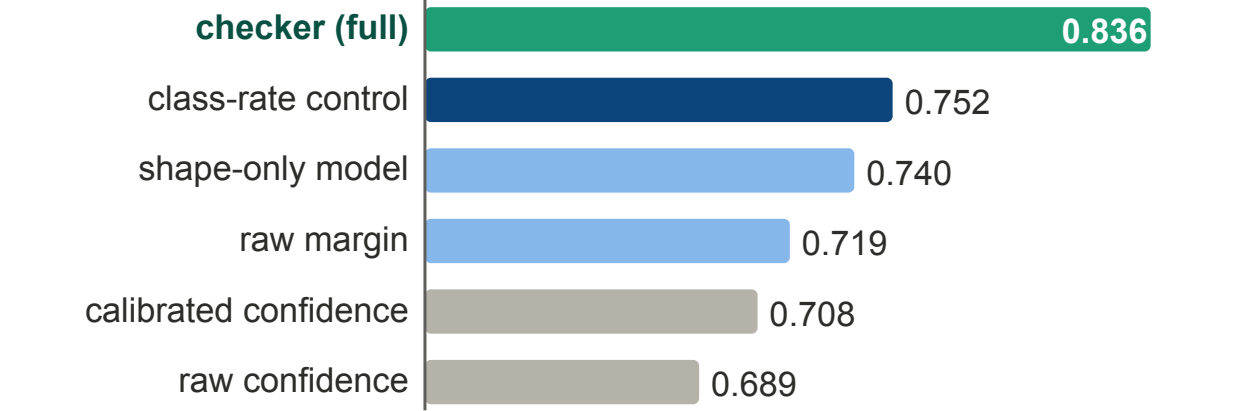
Policy @ 80% coverage	Acc.	Errors caught
Raw confidence	76.2%	34.5%
Predicted-class control	79.9%	44.5%
Checker (full vector)	81.0%	47.8%

...and not uniformly across its classes



Per-class ALeRCE ECE. SN Ia and SN Ibc are repaired; SN II degrades — the aggregate improves while one class gets worse.

The checker beats class identity alone



Δ = +0.083 over the class-rate control, paired-bootstrap 95% CI [0.057, 0.107] — object-level reliability beyond class identity. The shape-only model does not clear the control: the signal is in the full 15-class vector.

6 Conclusions

- Deployed systems miscalibrate in opposite directions — ALeRCE strongly underconfident, NEEDLE overconfident — so the correction a system needs cannot be read off its aggregate ECE.
- Calibration repairs what a probability means. It never changes the predicted class, but it decides which objects clear a fixed gate.
- One global correction is not enough: it degrades ALeRCE's SNII, and NEEDLE's own classes need temperatures on opposite sides of 1.
- A checker reading the full probability vector identifies unreliable predictions beyond predicted-class identity, concentrating errors into a small deferred queue.

Calibration fixes the scale. A reliability checker decides which predictions to trust. Neither is optional if broker outputs are to drive automated follow-up.

A functioning alert classifier is not automatically functioning decision infrastructure.

Main references

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